Tanay Kotian

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Education

University of California, Berkeley **Aug 2024 - May 2028**
Bachelor of Arts in Statistics, Bachelor of Arts in Data Science *Berkeley, CA*

Relevant Coursework

Foundations of Data Science (Data C8), Computational Structures of Data Science (Data C88C), Principles and Techniques of Data Science (Data 100), Linear Algebra and Differential Equations (Math 54), Calculus I and II (Math 51 and 52)

Experience

Interlinked **May 2025 - Aug 2025**
Artifical Intelligence and Data Engineer Intern *Berkeley, California*

- Analyzed and rigorously tested machine learning models (CNNs, neural networks, XGBoost) using LandFire datasets to enhance wildfire prediction.
- Authored and curated an in-depth master document synthesizing 10+ research papers, 5+ model architectures, 4+ algorithms (GOFER, PHOENIX Rapidfire, Rothermel algorithm, cWGAN), and experimental findings, providing strategic direction.
- Designed and evaluated physics-based and stochastic fire spread simulations, enabling dynamic, visually interpretable models for forecasting wildfire perimeters in real-world conditions.

Custex **June 2025 - Aug 2025**
Strategy and Software Engineer Intern *Berkeley, California*

- Designed and presented wireframes for a mobile-first personalized wellness and support platform dashboard, focusing on intuitive data visualization and user experience tailored for parental engagement.
- Developed a responsive React prototype using Tailwind CSS, converting wireframes into an interactive and user-friendly interface with simulated data integration.
- Conducted comprehensive technical due diligence on cross-device communication protocols, evaluating security, encryption standards, and network resilience to recommend best practices for secure data exchange.

Projects

Heart Attack Analysis | *Research Project Developer* **Jan 2025 - May 2025**

- Engineered a machine learning pipeline to predict heart attack risk using classification models such as Logistic Regression, Random Forest, and XGBoost.
- Conducted in-depth exploratory data analysis with Pandas, Matplotlib, and Seaborn to extract insights and uncover key risk factors.
- Optimized predictive accuracy to above 90% through hyperparameter tuning, cross-validation, and evaluation using precision and recall.

Bias In Predictive Policing | *Research Project Developer* **Sep 2024 - Dec 2024**

- Collaborated with 4+ peers aiming to address biases in predictive policing algorithms, with a focus on the COMPAS algorithm.
- Assisted with data analysis tasks such as data cleaning, feature engineering, and building models to evaluate potential discrimination in predictive policing.
- Analyzed the impact of predictive policing models on the criminal justice system—applying statistical and machine learning techniques to assess fairness, accuracy, and disproportionate effects on specific groups.

Pneumonia Imaging and Detection | *Research Student - InspiritAI* **April 2023 - June 2023**

- Led 8+ peers to fine-tune AI model to detect Pneumonia among patients through the usage of neural networks and testing data.
- Developed programming skills in AI via completion of Python coding notebooks and lessons taught by experts.
- Explored building LLM Agents and helped implement AI Apps in Gradio while working with professional Data Scientists.

Technical Skills

Languages: Python, SQL
Developer Tools: VS Code, Jupyter, Github
Libraries: XGBoost, Pandas, Matplotlib, Seaborn, Scikit-Learn